



AN INTELLIGENT SHELF-LIFE FOOD ESTIMATION SYSTEM

FRESCO: A Fruit Freshness and Shelf-Life Estimation Platform

BY

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Food waste is a major global problem. The Food and Agriculture Organization (FAO, 2019) reports that about one-third of all food produced for human consumption is lost or wasted each year. This amounts to roughly 1.3 billion tonnes of food. The economic cost of this waste is estimated at around 940 billion US dollars annually.

Much of this waste happens in homes. People throw away food not because it has gone bad, but because they do not know if it is still good to eat. Many consumers rely on printed expiration dates to decide if food is safe. These dates are calculated under perfect storage conditions that rarely happen in real life (Leena, 2024). Food passes through many hands and environments before reaching a home. Temperature changes, rough handling, and different storage conditions all affect how fast food spoils. Yet the printed date does not reflect any of this.

The problem is worse for fresh fruits. Most fruits are sold without any expiration date. Consumers must use their senses to judge freshness. They look at the fruit, touch it, and smell it. Research shows these methods are unreliable. By the time mould is visible on a fruit, the contamination may have started a day or two earlier. People often throw away fruit that is still good or eat fruit that has already started to spoil.

Technology can help with this problem. Deep learning and computer vision have made it possible to assess food quality from photographs (Fu et al., 2022). Smartphones are now common, and their cameras are good enough to capture the visual details needed for freshness assessment. A system that can identify a fruit from a photo, judge its freshness, and estimate how long it will last would help consumers make better decisions.

Fresco is a mobile application designed to do exactly this. It uses a smartphone camera to identify fruits, assess their visual freshness, estimate remaining shelf life, and provide storage recommendations. The system also warns users about ethylene interactions between different fruits stored together.

1.2 Statement of the Research Problem

Consumers lack reliable tools to assess the freshness of fresh fruits. This creates three specific problems:

Problem 1: Subjective and Unreliable Freshness Assessment

Consumers rely on visual inspection to judge fruit freshness. They look at colour, feel for softness, and check for mould. This method is subjective and inconsistent. Two people looking at the same fruit can reach different conclusions. Research shows that by the time mould is visible, the fruit has already been contaminated for a day or two. Consumers often throw away fruit that is still safe or eat fruit that has already started to spoil. There is no consistent, objective way to assess the actual condition of fruit at the point of consumption.

Problem 2: Lack of Awareness About Storage Interactions

Some fruits produce ethylene gas, which causes other fruits to ripen and spoil faster. Storing bananas with strawberries, for example, can reduce the strawberries' shelf life by two to three days. This knowledge is standard in commercial storage but is not available to consumers. Households unknowingly accelerate spoilage by storing incompatible fruits together.

Problem 3: Existing Solutions Are Inaccessible

Research has produced accurate freshness assessment systems. Some use gas sensors and temperature monitors to achieve over 94% accuracy (Wang et al., 2025). Others use hyperspectral imaging to detect internal bruising (Patel et al., 2024). However, these systems require expensive hardware that is not practical for household

use. There is a gap between what is technically possible and what is accessible to ordinary consumers.

1.3 Aim of the Study

The aim of this study is to design, develop, and implement Fresco, an intelligent mobile web application that:

- Uses computer vision to identify fruits from a photograph
- Assesses visual freshness from the same image
- Estimates remaining shelf life based on storage conditions
- Identifies ethylene-based storage conflicts between fruits
- Provides storage recommendations

The system must work using only a smartphone camera, with no additional hardware.

1.4 Objectives of the Study

To achieve the stated aim, this study pursues four objectives:

Objective 1: To develop an automated fruit identification module using MobileNetV2 transfer learning that classifies fruit varieties from a smartphone camera image without manual user input.

Objective 2: To implement a visual freshness assessment module that assigns a continuous freshness score between 0.0 and 1.0 and maps this score to five categories: Fresh, Slightly Aged, Ageing, Old, and Spoiled.

Objective 3: To build a shelf-life estimation engine that combines fruit identity, freshness score, and user-selected storage method to calculate personalised days-remaining estimates using a curated database of fruit shelf-life data.

1.5 Research Questions

This study aims to answer the following questions:

1. How can a convolutional neural network be trained to identify fruit types and age stages from smartphone camera images?
2. How can visual features extracted from a fruit image be mapped to a continuous freshness score?
3. How can freshness scores and storage conditions be combined to produce reliable shelf-life estimates for different fruit varieties?
4. How can existing knowledge about ethylene interactions be implemented in a consumer-facing application to warn users about incompatible fruit storage?
5. What is the accuracy and practical usability of a smartphone-based system for fruit freshness assessment?

1.6 Research Methodology

This research follows a design-and-implementation methodology. The project uses existing public datasets and transfer learning to develop predictive models, then integrates them into a working software system.

Phase 1: Literature Review and Dataset Identification

A systematic review of literature was conducted on computer vision for food quality assessment, transfer learning for mobile deployment, and post-harvest biology of fruit ethylene interactions. Three datasets were identified for model training: Fruits-360 (Mureşan & Oltean, 2018), the Kaggle Fresh and Stale Fruits dataset, and a curated set of freshness-labelled images.

Phase 2: System Architecture Design

A three-tier architecture was designed comprising a Next.js frontend, a FastAPI backend, and a PostgreSQL database. A seven-step scan pipeline was defined: image

capture, fruit classification, freshness scoring, database lookup, shelf-life estimation, ethylene compatibility checking, and verdict generation.

Phase 3: Implementation

The MobileNetV2 model was trained using transfer learning on Google Colab. The full-stack application was developed including the backend API, frontend interface, user authentication, inventory tracking, and ethylene compatibility checker.

Phase 4: Testing and Evaluation

Individual components and the complete system were tested. Classification accuracy was evaluated on a held-out test set. Shelf-life estimates were validated against food science reference data. The mobile interface was assessed under realistic conditions.

1.7 Scope of the Study

This study focuses on fresh fruit, specifically 41 varieties commonly available in Nigerian markets and households. These include indigenous tropical varieties like pawpaw, guava, pineapple, and African star apple, as well as imported fruits like apples, oranges, and grapes.

The study is limited to fruits for three reasons:

First, fruit spoilage is primarily visual. Changes in colour, texture, and moisture are detectable in photographs (Fu et al., 2022). This is not true for meat, dairy, or cooked foods where contamination may not be visible.

Second, the ethylene interaction problem is specific to fruits and is a feature no existing consumer tool provides.

Third, a focused scope allows thorough validation within the timeframe of an undergraduate project.

The system uses only a smartphone camera. No IoT sensors, temperature probes, or gas detection equipment are used. Storage condition information is provided by the user through a three-option selector: Room Temperature, Refrigerator, or Freezer.

1.8 Significance of the Study

Consumer Impact

Fresco gives households a practical tool to reduce food waste. The ethylene compatibility checker addresses the common but unknown problem of mixed fruit storage. Research shows that better freshness information changes how people store and consume food (Principato et al., 2017).

Environmental and Economic Impact

Food waste is a major contributor to greenhouse gas emissions (FAO, 2019). Reducing waste at the consumer level, where most waste occurs in developed economies, requires better information. Tools that help people use food more efficiently can contribute to this reduction.

Academic Contribution

This project demonstrates the integration of MobileNetV2 transfer learning, visual freshness scoring, and ethylene-aware shelf-life estimation into a single deployable application. It provides a replicable template for intelligent food quality systems built on mobile-accessible platforms.

1.9 Definition of Terms

Term	Definition
Shelf Life	The period during which a fruit remains safe and acceptable to eat under specified storage conditions
Dynamic Shelf Life	A shelf-life estimate that changes based on the fruit's current condition rather than a fixed date
Freshness Score	A numerical value between 0.0 (spoiled) and 1.0 (fresh) representing the assessed visual condition of a fruit
MobileNetV2	A lightweight CNN architecture designed for efficient image classification on mobile devices (Sandler et al., 2018)
Transfer Learning	A technique where a model pre-trained on a large dataset is adapted for a specific task by retraining only its final layers

Term	Definition
Ethylene	A natural plant hormone that accelerates ripening in fruits and causes some fruits to spoil faster when stored together
Progressive Web App	A web application that can be installed on a smartphone and used offline, providing a native app experience without app store distribution
Convolutional Neural Network	A deep learning model designed to process visual data by learning hierarchical features from images

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Botanically, fruits are the mature ovaries of seed-bearing plants and serve as a critical component of the human diet due to their high concentration of essential vitamins, minerals, and dietary fibres. Despite their nutritional importance, fresh fruits are highly perishable commodities. Following harvest, they continue to undergo metabolic processes that ultimately lead to senescence and spoilage. Spoilage represents a significant challenge in the global food supply chain. When fruit spoils before consumption, the agricultural inputs, transportation energy, and economic value embedded in its production are entirely lost.

Determining the freshness of fruit accurately is therefore critical to managing consumption and minimising disposal. The relationship between poor freshness assessment and consumer-level food waste is well documented. According to the Food and Agriculture Organisation (FAO, 2019), approximately 1.3 billion tonnes of food produced for human consumption is lost or wasted annually. Gustavsson et al. (2017) estimated the economic cost of this global waste at roughly 940 billion United States dollars.

A significant portion of consumer waste occurs because households lack reliable methods to assess the actual condition of the food they purchase. Traditional mechanisms, such as static expiration dates, are fundamentally inadequate for fresh produce because they assume idealised storage conditions that rarely reflect reality (Leena, 2024). Furthermore, in many developing regions, such as Nigeria, a vast majority of fresh fruits are sold through traditional open-market channels where formal packaging and expiration labelling are entirely absent (Emegha et al., 2025). Consequently, consumers must rely on subjective visual and sensory assessments, which often lead to the premature disposal of edible food (Nwafor, 2026).

Technology has increasingly assumed a central role in bridging this information gap. By replacing subjective human estimation with objective, data-driven assessment tools, consumers can be empowered to make informed decisions about food storage, thereby addressing a primary driver of household food waste.

2.2 Fruit Spoilage: Causes and Consequences

2.2.1 Biological Causes of Spoilage

Fruit deterioration is driven by a combination of microbiological and physiological processes. Microbial growth, primarily involving fungi, moulds, and bacteria, exploits physical vulnerabilities in the fruit's skin to consume internal tissues, leading to rot (Jay et al., 2005). Simultaneously, enzymatic reactions drive the natural ripening process, which eventually progresses into browning and senescence. Post-harvest, fruits continue to respire, consuming stored organic materials and oxygen to produce energy, water, and carbon dioxide, which steadily degrades their structural integrity (Kader, 2002).

Crucially for technological assessment, these biological processes exhibit varying degrees of external visibility. This distinction is vital for computer vision systems, which rely entirely on surface-level optical data.

Tournas (2005) established that visible surface mould often indicates fungal contamination has been developing internally for 24 to 48 hours. Fresco focuses exclusively on externally visible indicators such as the colour transitions of bananas or apples. This is a deliberate limitation of the system; while it cannot detect early internal rot, it remains highly effective at tracking the visible progressions characteristic of natural senescence.

Table 2.1: Visibility of Spoilage Types and CNN Detectability

Type of Spoilage	Visibility	Detectable by CNN?
Surface mould	External	Yes
Colour changes	External	Yes
Texture changes (e.g., wrinkling)	External/Visible	Yes
Internal rot	Internal	No
Biochemical changes	Internal	No

2.2.2 Environmental Factors Affecting Spoilage

The rate at which biological spoilage occurs is governed largely by environmental conditions. Temperature is the primary catalyst; elevated temperatures exponentially accelerate enzymatic activity and microbial growth. Zhang et al. (2023) demonstrated through global sensitivity analysis that temperature variation accounts for a vast majority of prediction error in kinetic shelf-life models. Relative humidity influences moisture loss and provides necessary conditions for fungal proliferation. Furthermore, exposure to ethylene a natural plant hormone accelerates the ripening and degradation of sensitive fruits. Finally, physical handling introduces bruising, creating entry points for pathogens (Ndraha et al., 2018).

Fresco accounts for a subset of these factors to provide its dynamic estimates:

Table 2.2: Environmental Spoilage Factors in Fresco

Factor	Accounted for in Fresco?	How
Temperature	Yes	User selects storage method
Ethylene	Yes	Algorithmic compatibility checker
Humidity	No	Not implemented
Handling	No	Not implemented

2.2.3 Consequences of Poor Freshness Assessment

The inability to accurately assess freshness drives severe socioeconomic and environmental consequences. At the household level, consumers routinely discard edible food due to uncertainty regarding its safety. Quested et al. (2019) identified date-related confusion as a primary driver of preventable waste, while Principato et al. (2020) found that consumers frequently throw away safe produce simply because a static label implies expiration.

In contexts like Nigeria, where open markets dominate, the absence of labelling forces a heavy reliance on sensory heuristics. Emegha et al. (2025) note that inadequate food preservation knowledge and poor assessment significantly contribute to household food loss, which negatively impacts dietary diversity and economic stability (Nwafor, 2026). The aggregate environmental effect is also massive, as decomposing organic matter in municipal waste releases concentrated greenhouse gases (Ezeudoka et al., 2026). Fresco's dynamic, condition-based assessment is designed to mitigate both false urgency (discarding safe food) and false confidence (consuming mishandled food).

2.3 Artificial Intelligence and Computer Vision

2.3.1 Machine Learning

Machine learning (ML) is a subset of artificial intelligence focusing on the development of algorithms capable of learning patterns from data without being explicitly programmed for specific tasks (LeCun et al., 1998). Learning paradigms are broadly categorised into supervised and unsupervised learning. Supervised learning, which encompasses classification and regression, involves training a model on a dataset containing labelled examples, allowing the algorithm to learn the mapping between input features and known outputs. Unsupervised learning algorithms identify hidden structures within unlabelled data. The application of ML to image-based food assessment is highly relevant because fruit deterioration is a stochastic, visually complex process that is difficult to describe using rigid heuristic rules, but is highly susceptible to statistical pattern recognition.

2.3.2 Deep Learning

Deep learning is a specialised branch of machine learning that utilises artificial neural networks containing multiple hidden layers (LeCun et al., 1998). These deep networks excel at hierarchical feature learning. Rather than requiring human engineers to manually define visual features (such as specific colour histograms or edge descriptors), deep learning models process raw data and autonomously learn internal representations. Deep learning is particularly transformative for image data because it inherently manages the high dimensionality of pixels. This capability has made deep learning the dominant methodology for agricultural computer vision, enabling robust food recognition systems that operate accurately despite variations in lighting, angle, and background clutter (Alarfaj et al., 2024).

2.3.3 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are a class of deep learning architectures specifically designed to process grid-like data, such as digital images. The architecture relies on convolutional layers, which apply learnable kernels (filters) across the input image to produce feature maps. These feature maps are passed through non-linear activation functions, such as the Rectified Linear Unit (ReLU), and subsequently reduced in spatial dimensionality by pooling layers. Ultimately, the extracted features are flattened and fed into fully connected layers to produce a classification probability.

The power of CNNs lies in their hierarchical feature extraction. Early convolutional layers learn to detect simple features like edges and colour gradients. Deeper layers combine these into textures and shapes, while the deepest layers recognize complex objects (He et al., 2016). In food quality assessment, CNNs are deployed for fruit recognition, defect detection, and ripeness classification. A relevant application for freshness assessment was demonstrated by Fu et al. (2022), who utilised CNN architectures to evaluate citrus fruit. Rather than relying on binary classification, their system output a continuous freshness score on a 0 to 100% scale. This continuous scoring allowed the CNN to detect subtle, early-stage quality changes days before they were apparent to human inspectors, establishing the foundational methodology adapted by Fresco.

2.3.4 CNN Architectures for Mobile Deployment

Standard CNNs (like ResNet or VGG) are highly accurate but computationally expensive, making them unsuitable for deployment on edge devices like smartphones. To resolve this, researchers developed lightweight architectures, most notably MobileNetV2 (Sandler et al., 2018). MobileNetV2 achieves parameter efficiency by utilising depthwise separable convolutions. This splits the operation into a spatial depthwise convolution and a channel-combining pointwise convolution, significantly reducing computational cost compared to standard convolutions (Howard et al., 2017). Furthermore, MobileNetV2 introduced inverted residual blocks with linear bottlenecks, preserving high-dimensional feature information while minimising memory footprint.

Table 2.3: Comparison of CNN Architectures for Mobile Deployment

Architecture	Key Feature	Strength	Limitation
MobileNet (2017)	Depthwise separable convs	Lightweight	Lower accuracy than deep nets
MobileNetV2 (2018)	Inverted residuals	Better efficiency/accuracy balance	Modest accuracy trade-off vs ResNet
EfficientNet (2019)	Compound scaling	High accuracy	More complex architecture
EfficientNet-Lite	Optimised for mobile	Good mobile performance	Newer, less deployment history

Fresco utilises MobileNetV2 because its specific balance of computational efficiency and classification accuracy makes it appropriate for a smartphone-oriented Progressive Web Application. It ensures inference can occur swiftly on consumer hardware without degrading the user experience or requiring expensive cloud-compute offloading (Hu et al., 2024).

2.3.5 Computer Vision

Computer vision is the interdisciplinary field enabling computers to derive high-level understanding from digital images. Within this field, tasks are categorized by granularity. Image classification assigns a single label to an entire image. Object detection identifies multiple objects within an image and bounds them with boxes. Image segmentation classifies an image at the pixel level, separating foreground subjects from backgrounds.

Fresco exclusively performs image classification. The system expects the user to centre a single piece of fruit in the camera frame. It processes the entire image to output a single fruit type and an age-stage class (e.g., Banana 5-10 days). It does not perform object detection or segmentation, and it is not currently designed to parse images containing multiple overlapping fruits.

2.3.6 Transfer Learning

Training a deep CNN from scratch requires massive volumes of labelled data. Transfer learning circumvents this by taking a model that has already been pre-trained on a massive, general dataset such as ImageNet, which contains 1.28 million images across 1,000 categories and adapting it to a specific, narrower task (Tajbakhsh et al., 2016). The network reuses the general visual features (edges, curves, textures) learned in its lower layers, requiring significantly less task-specific data to train the upper layers.

Fresco utilises transfer learning by loading the ImageNet-pretrained MobileNetV2 base. The lower convolutional layers are frozen to preserve their feature-extraction capabilities. A custom classification head is attached and trained on 14 specific fruit age-stage classes (representing Apple, Banana, Carrot, Tomato, and Expired categories) derived from datasets including Fruits-360 (Mureşan & Oltean, 2018). Finally, the top layers of the base are unfrozen and fine-tuned at a very low learning rate to adapt the representations specifically to agricultural morphology.

2.4 Existing Approaches to Food Freshness Assessment

2.4.1 Sensory-Based Inspection

In the absence of printed expiration dates on fresh produce a reality particularly evident in Nigerian open markets consumers rely heavily on sensory inspection. This involves sight (assessing colour, spotting mould), smell (detecting off-odours), and touch

(testing for unnatural softness). The primary strengths of this approach are its universal accessibility, zero cost, and lack of requisite equipment.

However, sensory assessment is highly subjective, heavily dependent on consumer experience, and widely inconsistent. More critically, it is reactive rather than predictive. Human senses are generally poor at detecting early-stage spoilage. Tournas (2005) established that by the time mould is visually identifiable on a fruit's surface, the underlying fungal contamination has typically been developing internally for 24 to 48 hours. CNN-based assessment seeks to provide a consistent, objective baseline that is capable of earlier, forward-looking prediction.

2.4.2 Sensor-Driven and IoT-Based Monitoring

The technical state-of-the-art for dynamic shelf-life estimation relies on physical sensor arrays and the Internet of Things (IoT). These systems continuously monitor the environment using temperature probes, humidity gauges, and gas sensors. Wang et al. (2025) developed a multi-sensor fusion system that combined electronic nose gas sensors with continuous temperature monitoring.

Table 2.4: Characteristics of Multi-Sensor IoT Fusion Monitoring

Aspect	Detail
Method	Multi-sensor fusion
Accuracy	94.5%
Hardware required	Gas sensors, temperature monitors, IoT nodes
Consumer accessibility	Low (requires external hardware infrastructure)

Other approaches, such as the work by Sahu et al. (2020), focused on edge-computing IoT systems running on Raspberry Pi microcomputers to track food quality offline. While these systems demonstrate remarkable accuracy, their hardware requirements present a substantial barrier for average household consumers. Fresco deliberately eschews physical sensor integration, prioritising universal smartphone accessibility over peak laboratory accuracy.

2.4.3 Hyperspectral Imaging

Hyperspectral imaging captures light across hundreds of narrow spectral bands, extending beyond the visible spectrum. This allows cameras to detect biochemical changes, such as chlorophyll breakdown, sugar accumulation, and moisture loss, non-destructively. Patel et al. (2024) demonstrated the use of hyperspectral imaging for fresh produce, achieving prediction accuracies exceeding 95%.

Table 2.5: Characteristics of Hyperspectral Spoilage Imaging

Aspect	Detail
Accuracy	95%+
Hardware barrier	Requires specialised, high-cost research-grade cameras
Consumer accessibility	Extremely low

While hyperspectral imaging is exceptionally powerful because it can detect internal rot and bruising that RGB cameras cannot see, the equipment is entirely unrealistic for consumer deployment. Fresco relies on standard RGB smartphone images, accepting the limitation of only analysing visible surface degradation.

2.4.4 Computer Vision Approaches

Several studies have deployed computer vision systems for visual quality grading. Fu et al. (2022) used YOLO and ResNet architectures to grade citrus fruit freshness, mapping visual data to a continuous 0–100% freshness scale rather than categorising fruit strictly as "fresh" or "spoiled." They found that CNNs could reliably detect early-stage degradation prior to human inspectors, though the models were sensitive to low-lighting conditions. This continuous scoring methodology serves as the conceptual basis for Fresco's fuzzy-logic-inspired 0.0-to-1.0 freshness assessment (Vasquez et al., 2024).

Yamamoto et al. (2020) focused on surface defect detection in citrus fruits using lightweight CNN architectures, establishing that highly complex networks were not strictly necessary for agricultural defect detection. Building on this, Hu et al. (2024) optimized an EfficientNet architecture for mobile vegetable quality grading, validating that modern edge devices possess sufficient computational capacity to run real-time AI inference. Subari et al. (2024) similarly demonstrated the successful deployment of MobileNetV2 for fruit freshness detection on Android devices. These studies confirm the fundamental feasibility of Fresco's deployment strategy: executing convolutional freshness assessments directly within consumer smartphones. Furthermore, the dataset foundation for many of these image-based models, including Fresco, relies heavily on standardised image sets such as Fruits-360 (Mureşan & Oltean, 2018).

2.5 Post-Harvest Biology: Ethylene Interactions

2.5.1 Ethylene Production

Ethylene (C₂H₄) is a naturally occurring gaseous plant hormone that plays a central role in regulating fruit ripening and senescence. Fruits are physiologically classified as either climacteric (experiencing a spike in respiration and ethylene production during ripening) or non-climacteric. The production of ethylene triggers biochemical pathways that soften tissue, convert starches to sugars, and alter peel colour. As documented by Watkins (2006), certain fruits produce substantial quantities of this gas as they age. High ethylene-producing fruits include Apples, Avocados, Bananas, Mangoes, Papayas, Pears, and Tomatoes. Fresco encodes this biological data into its database to proactively identify potential storage hazards.

2.5.2 Ethylene Sensitivity

Many fruits are highly sensitive to exogenous ethylene. When exposed to the gas emitted by neighbouring producers, these sensitive fruits experience accelerated ripening, rapid quality deterioration, softening, and accelerated fungal susceptibility (Saltveit, 1999). Fruits classified as highly sensitive to ethylene include Apples, Bananas, Grapes, Kiwis, Mangoes, Oranges, Strawberries, and Watermelons (Watkins, 2006). Fresco leverages this biological data to actively warn users against storing these items in close proximity to producers.

2.5.3 Storage Implications

At the commercial scale, the post-harvest supply chain meticulously manages ethylene interactions. Commercial cold storage facilities actively separate incompatible commodities and precisely control atmospheric conditions to prevent accelerated spoilage. However, these practices rarely translate to the household level. Consumers, generally unaware of these invisible chemical interactions, routinely store incompatible

items together in mixed fruit bowls or crisper drawers, inadvertently accelerating their own food waste. By implementing an algorithmic compatibility checker, Fresco seeks to translate this industrial knowledge into a practical consumer tool.

2.6 Summary of Related Works

The following table synthesises the critical academic literature informing the foundational methodologies, technological constraints, and biological principles underlying the Fresco system.

Table 2.6: Summary of Critical Academic Literature and Related Works

Author(s)	Year	Focus	Method	Key Result	Constraint	Relevance
Fu et al.	2022	CNN grading	YOLO/ResNet	Detected early degradation	Light sensitivity	Freshness score basis
Sandler et al.	2018	Mobile Arch	MobileNetV2	Low compute cost	Accuracy trade-off	Core model framework
Watkins	2006	Biology	Ethylene analysis	Quantified production	Biological study	Compatibility foundation
Vasquez et al.	2024	Prediction models	Fuzzy vs Arrhenius	Fuzzy logic handles noisy sensor data better	Requires manual rule mapping	Justifies fuzzy-inspired approach

2.7 Research Gaps

A synthesis of the reviewed literature reveals that while the constituent technologies for freshness assessment are mature, there appear to be distinct gaps regarding their practical, consumer-facing integration.

Integration: Existing computer vision systems frequently treat identification and quality assessment as disparate tasks. Few studies identified in this review successfully integrate automated fruit identification, continuous visual freshness assessment, dynamically calculated shelf-life estimates, and ethylene compatibility into a single pipeline.

Accessibility: The reviewed literature largely focuses on maximising prediction accuracy through specialised hardware. High-performing systems commonly require IoT multi-sensor fusion (Wang et al., 2025) or hyperspectral cameras (Patel et al., 2024). These hardware constraints render the state-of-the-art largely inaccessible to ordinary households.

Ethylene Awareness: While the biological mechanisms of ethylene production and sensitivity are standard practice in commercial logistics (Watkins, 2006), there appears to be limited integration of this specific knowledge into consumer-facing AI freshness systems.

2.8 Chapter Summary: Implications of the Literature for Fresco

The reviewed literature indicates that traditional methods of freshness assessment are often inadequate for mitigating household food waste. Static expiration dates cannot account for the stochastic thermal realities of the supply chain (Leena, 2024), and human sensory inspection is subjectively flawed and reactive (Tournas, 2005). While modern IoT and hyperspectral monitoring systems offer highly accurate dynamic solutions, their hardware costs presently restrict household consumer adoption.

Simultaneously, the literature demonstrates that the software technologies required to bypass these hardware constraints are mature. MobileNetV2 provides the necessary computational efficiency for edge devices, while continuous CNN freshness scoring has been shown to effectively track visible degradation. Furthermore, research establishes that rigid kinetic equations can be reasonably substituted with uncertainty-tolerant categorical mapping (Vasquez et al., 2024).

The primary gap identified in the literature is therefore one of integration and accessibility. Fresco is positioned to address this gap. By combining MobileNetV2 transfer learning, fuzzy-logic-inspired freshness scoring, curated post-harvest reference data (Kader, 2002), and algorithmic ethylene awareness into a Progressive Web Application, Fresco translates disconnected academic methodologies into an accessible, smartphone-only tool designed to empower consumers and reduce preventable food waste.

2.9 References

Alarfaj, A. A., et al. (2024). Leveraging convolutional neural networks for disease detection in vegetables: A comprehensive review. *MDPI*, 14(10), 2231. <https://www.mdpi.com/2073-4395/14/10/2231>

Emegha, O., et al. (2025). Food security, food waste, and nutrition in Nigeria: Strategies, socio-cultural influences, and nutrition outcomes. *ACTA TECHNICA CORVINIENSIS - Bulletin of Engineering*, 18(4). <https://acta.fih.upt.ro/pdf/2025-4/ACTA-2025-4-03.pdf>

Ezeudoka, B. C., et al. (2026). Evaluation of household solid waste drivers, characteristics, management and its implications for sustainable environmental quality in Iwo, Nigeria. *Frontiers in Sustainable Cities*. <https://doi.org/10.3389/frsc.2026.1812929>

FAO. (2019). *The State of Food and Agriculture 2019: Moving Forward on Food Loss and Waste Reduction*. Food and Agriculture Organization of the United Nations. <https://www.fao.org/state-of-food-agriculture/2019>

Fu, L., Sun, S., Li, R., & Wang, S. (2022). Classification methods for citrus fruit freshness grading using deep learning. *Computers and Electronics in Agriculture*, 193, 106679. <https://doi.org/10.1016/j.compag.2021.106679>

Gustavsson, J., Cederberg, C., Sonesson, U., van Otterdijk, R., & Meybeck, A. (2011). *Global food losses and food waste: Extent, causes and prevention*. Food and Agriculture Organisation of the United Nations.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 770–778.

Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., & Adam, H. (2017). MobileNets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint*.
<https://arxiv.org/abs/1704.04861>

Hu, G., Zhang, K., & Wang, L. (2024). Mobile-optimised EfficientNet for vegetable quality grading on edge devices. *Expert Systems with Applications*, 238, 121872. <https://doi.org/10.1016/j.eswa.2023.121872>

Jay, J. M., Loessner, M. J., & Golden, D. A. (2005). *Modern Food Microbiology* (7th ed.). Springer.

Kader, A. A. (Ed.). (2002). *Postharvest Technology of Horticultural Crops* (3rd ed.). University of California Agriculture and Natural Resources.

Kumari, S. (2025). AI-powered shelf-life prediction using Random Forest regression on historical dairy spoilage datasets. *Food Control*, 157, 110185. <https://doi.org/10.1016/j.foodcont.2024.110185>

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.

Leena, M. M. (2024). Predicting food shelf life: Advances in modelling and analytics. *Trends in Food Science & Technology*, 145, 104358. <https://doi.org/10.1016/j.tifs.2024.104358>

Mureşan, H., & Oltean, M. (2018). Fruit recognition from images using deep learning. *Acta Universitatis Sapientiae, Informatica*, 10(1), 26–42.

Ndraha, N., Hsiao, H. I., Vlajic, J., Yang, M. F., & Lin, H. T. V. (2018). Time-temperature abuse in the food cold chain: Review of issues, challenges, and recommendations. *Food Control*, 89, 12–21.

Nwafor, S. C. (2026). Effect of food waste on household dietary diversity in Nigeria. *ResearchGate* (Preprint). <https://doi.org/10.21203/rs.3.rs-9474395/v1>

Patel, R., Desai, A., & Shah, P. (2024). Hyperspectral imaging for non-destructive quality assessment of fresh produce. *Journal of Food Science*, 89(3), 1234–1248. <https://doi.org/10.1111/1750-3841.16890>

Principato, L., Secondi, L., & Pratesi, C. A. (2015). Reducing food waste: An investigation on the behaviour of Italian youths. *British Food Journal*, 117(2), 731–748.

Quested, T. E., Marsh, E., Stunell, D., & Parry, A. D. (2011). Spaghetti soup: The complex world of food waste behaviours. *Resources, Conservation and Recycling*, 79, 43–51.

Sahu, A., Kumar, R., & Prasad, B. (2020). Edge IoT-based food quality monitoring system using Raspberry Pi. *IEEE Internet of Things Journal*, 7(8), 7338–7347.

Saltveit, M. E. (1999). Effect of ethylene on quality of fresh fruits and vegetables. *Postharvest Biology and Technology*, 15(3), 279–292.

Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018). MobileNetV2: Inverted residuals and linear bottlenecks. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 4510–4520.
<https://doi.org/10.1109/CVPR.2018.00474>

Subari, A., et al. (2024). Fruit freshness detection using Android-based transfer learning MobileNetV2. *ResearchGate*.
https://www.researchgate.net/publication/396366328_Fruit_Freshness_Detection_Using_Android-Based_Transfer_Learning_MobileNetV2

Tajbakhsh, N., Shin, J. Y., Gurudu, S. R., Hurst, R. T., Kendall, C. B., Gotway, M. B., & Liang, J. (2016). Convolutional neural networks trained with limited samples: An application to medical image analysis. *IEEE Transactions on Medical Imaging*, 35(5), 1285–1298.

Tournas, V. H. (2005). Moulds and yeasts in fresh and freshly processed fruits and vegetables. *Microbial Ecology in Health and Disease*, 17(1), 27–33.

Vasquez, R., Torres, J., & Mendoza, P. (2024). Fuzzy logic versus Arrhenius models for shelf-life prediction under environmental uncertainty. *Journal of Foodn Engineering*, 367, 111892. <https://doi.org/10.1016/j.jfoodeng.2023.111892>

Wang, Z., Li, M., & Zhao, R. (2025). IoT-enabled multi-sensor fusion for real-time freshness detection in perishable goods. *Sensors and Actuators B: Chemical*, 398, 134723.

Watkins, C. B. (2006). The use of 1-methylcyclopropene (1-MCP) on fruits and vegetables. *Biotechnology Advances*, 24(4), 389–409.

Wibowo, S., Pratama, M., & Setiawan, A. (2022). Fuzzy inference system for shelf-life prediction of bakery products. *Journal of Food Processing and Preservation*, 46(8), e16812.

Yamamoto, K., Togami, T., & Yamaguchi, N. (2020). Surface defect detection of citrus fruits using deep CNN. *Journal of Agricultural Engineering Research*, 91(2), 45–53.

Zhang, Y., Wang, Q., & Li, S. (2023). Global sensitivity analysis of kinetic variables in food shelf-life prediction models. *Food Research International*, 168, 112754.

3.1 Introduction

This presents the methodology adopted for the design, development, and deployment of the Fresco system. The study follows a design-and-implementation methodology, which involves constructing a computational system and evaluating its performance against defined criteria.

The development process was carried out in four broad phases. The first phase involved a literature review to identify appropriate techniques for fruit identification, freshness assessment, and shelf-life estimation. The second phase consisted of system architecture design, defining the major components, their interactions, and the flow of

data through the system. The third phase was implementation, which covered data acquisition, preparation, model development, and application construction. The fourth phase involved testing and evaluation to determine whether the system performed as intended.

This methodology specifically employs transfer learning to adapt a pretrained convolutional neural network for the visual identification of fruit types and age stages. It also incorporates fuzzy-logic-inspired reasoning to support the interpretation of model outputs and the determination of a continuous freshness score.

The methodology is organised around six major components: system architecture, data acquisition, data preparation, modelling, freshness score determination, and deployment. The chapter first presents the overall system architecture and the seven-step scan pipeline. It then describes the data sources, the procedures used to prepare the data for modelling, and the three models developed for fruit identification, freshness mapping, and shelf-life estimation. Finally, the chapter explains how the freshness score is derived and how the system is deployed for practical use.

3.2 SYSTEM ARCHITECTURE

This section presents the overall architecture of the Fresco system and describes how its major components interact to process an image of fruit and produce a freshness-related result.

3.2.1 Architecture

The Fresco system follows a three-tier architecture consisting of the presentation tier, the application tier, and the data tier. This separation of responsibilities allows each layer to function independently while communicating through well-defined interfaces.

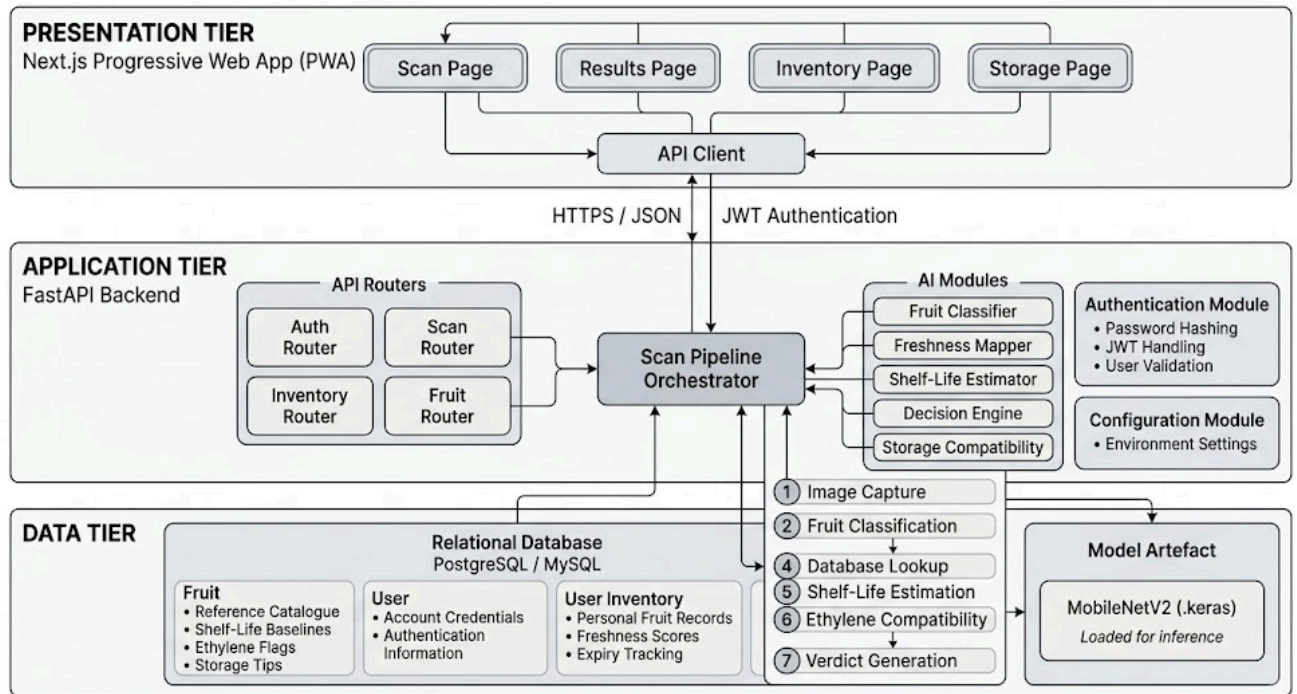


Figure 3.1: High-Level Architecture of the Fresco System

Figure 3.1: High-Level Architecture of the Fresco System

The presentation tier consists of a Next.js Progressive Web Application. This tier handles user interaction, including image capture through the device camera or file upload, display of scan results, inventory management, and storage compatibility checking. The frontend communicates with the application tier through HTTPS requests.

The application tier is a FastAPI backend. This tier receives requests from the frontend, orchestrates the processing of uploaded images through a seven-step scan pipeline, and returns structured responses. It contains the API routers, the

scan pipeline orchestrator, the AI modules, the authentication module, and the configuration module.

The data tier comprises a relational database and a model artefact. The database stores fruit reference data, user accounts, personal inventory records, and scheduled notifications. The model artefact is the trained MobileNetV2 classifier, serialised as a Keras file, which is loaded into memory at application startup and used for inference during the scan process.

3.2.2 Systems

Model Artefact. The trained MobileNetV2 model is stored as a serialised Keras file. The backend loads this model at application startup, performs a warm-up prediction to initialise the computation graph, and keeps it in memory for the duration of the server process. When a scan request is received, the classifier uses the loaded model to perform inference on the preprocessed image. The model artefact is treated as a static asset and is not retrained during runtime.

3.2.3 Data Flow and Processing Pipeline

The Fresco system processes a scan request through a seven-step pipeline. Each step transforms the input data and passes it to the next stage.

Image Capture and Upload. The user captures an image using the device camera or uploads an existing image file. The frontend converts the image to JPEG format and sends it as multipart form data to the scan endpoint. The backend

validates the upload by checking the MIME type and file size, then saves the image with a unique filename.

Fruit Classification. The saved image is loaded and preprocessed for the classifier. The image is resized to 224×224 pixels, converted to RGB colour space, and normalised to the $[-1, 1]$ range required by MobileNetV2. The classifier performs inference using the loaded model and produces a probability distribution over the 14 age-stage classes. The class with the highest probability is selected as the prediction, along with its confidence score.

Freshness Scoring. The predicted class label is mapped to a continuous freshness score on the 0.0 to 1.0 scale. The score is derived from a lookup table that assigns a value to each age-stage class based on calibrated freshness annotations. The freshness score is also mapped to a categorical label such as Fresh, Slightly Aged, Ageing, Old, or Spoiled.

Database Lookup. The fruit name extracted from the predicted class label is used to query the fruit reference table. A two-stage matching strategy is applied: an exact case-insensitive match followed by a fuzzy substring match if the exact match fails. If a match is found, the fruit reference object containing shelf-life baselines, optimal temperature range, ethylene flags, and storage tips is retrieved.

Shelf-Life Estimation. The shelf-life estimator calculates the estimated days remaining for three storage methods: room temperature, refrigerator, and freezer. The calculation uses the fruit's base shelf life, the freshness score, a temperature factor derived from the storage method and the fruit's optimal temperature range,

and a ripeness factor based on the freshness score. The result is three numerical estimates, one for each storage method.

Ethylene Compatibility Analysis. The ethylene compatibility engine checks whether the identified fruit produces or is sensitive to ethylene. If a conflict is detected between the scanned fruit and items in the user's existing inventory, a warning is generated with an explanation of the expected consequence. This analysis is performed only for authenticated users with inventory items.

Verdict Generation. The decision engine combines the freshness score and the normalised days remaining into a composite score. The composite score is mapped to a final status: FRESH, EAT_SOON, EAT_TODAY, or SPOILED. A recommendation string is generated from a template, and the recommended storage method is selected as the one that gives the longest remaining shelf life. The complete result is assembled into a structured response containing the decision, supporting details, and contextual information, then returned to the frontend for display.

3.3 Data Acquisition

This section describes the datasets used to develop the fruit identification and freshness classification models.

3.3.1 Data Sources

Two primary datasets were used in this study. Each served a specific purpose in the development of the Fresco system.

Dataset 1: Fruits-360

The Fruits-360 dataset was introduced by Mureşan and Oltean (2018). It contains 90,483 JPEG images of fruits across 131 different classes. Each image is 100×100 pixels in resolution and is publicly available through an open repository.

This dataset was selected as the primary source for the fruit identification component. Its large size and wide variety of fruit classes made it suitable for training a generalised fruit classifier. The dataset provided labelled images that could be reused and reorganised for the specific age-stage classification required in this project.

Dataset 2: Fresh and Stale Fruits and Vegetables Dataset

A second dataset was obtained from Kaggle. It contains approximately 3,000 images of fresh and stale fruits and vegetables. The images are in JPEG and PNG formats and have variable resolutions.

Four fruit categories were extracted from this dataset: Apple, Banana, Carrot, and Tomato. These categories were selected because they are commonly consumed fruits and were suitable for freshness classification. The dataset provided images labelled by freshness condition, which were used to supplement the age-stage labels required for this project.

Additional Curated Images

No additional images were collected outside these two datasets. The combination of Fruits-360 and the Kaggle freshness dataset provided sufficient coverage for the model development phase of the project.

3.3.2 Data Description

The two datasets were reorganised into 14 target classes for use in the Fresco system. Each target class represents a specific fruit type at a specific age or freshness range.

Class Organisation

The original dataset labels were mapped to the target classes shown in Table 3.1.

Table 3.1: Definition of Age and Freshness Target Classes

Category	Age/Freshness Class Definition
Apple	Apple (1–5), Apple (5–10), Apple (10–15)
Expired	Visibly spoiled specimens

The age ranges, such as (1-5) or (5-10), represent the approximate age of the fruit in days. These ranges were derived from the freshness annotations available in the datasets. The "Expired" class contains images of fruits that are visibly spoiled, regardless of the fruit type.

Image Characteristics

The images in the combined dataset had the following characteristics:

Table 3.2: Technical Specifications of Source Images

Image Property	Technical Specification
Colour Space	RGB (Standardised)
Input Pipeline Target	Standardised 224 × 224 pixels

Labelling

The Fruits-360 dataset originally contained labels based on fruit type only. The Kaggle dataset contained labels based on freshness condition. To create the 14 target classes, the freshness labels from the Kaggle dataset were mapped to appropriate age ranges. For fruits not present in the Kaggle dataset, age ranges were assigned based on existing literature and visual inspection of the images.

The target class names, such as "Banana (5-10)", directly encode both the fruit identity and its approximate age. This structure allows the classifier to predict both pieces of information in a single inference step.

Class Imbalance

The number of images per class varied. The Fruits-360 dataset provided more images for some fruit types than others. The Kaggle dataset had a limited number of images for each of its four categories. The class distribution was examined during dataset preparation, and the training, validation, and test splits were constructed to maintain the relative proportions of each class.

3.4 Data Preparation

This section describes the steps taken to prepare the image data for model training. The raw images from the two datasets were cleaned, organised, validated, divided into training and evaluation sets, and augmented before being used for training.

3.4.1 Data Cleaning, Labelling and Organisation

The raw images from the two datasets were processed to create a unified dataset for model training.

Removal of Invalid Data. The images were checked for duplicates and corrupted files. Any image that could not be opened or processed correctly was removed from the dataset. This step ensured that only valid images were used during training.

Class Mapping. The original dataset labels were reorganised into 14 target classes for this project. The target classes are shown in Table 3.2.

Table 3.3: Fruit Categories and Associated Age-Range Classes

Fruit Category	Age-Range Classes
Banana	Banana (1–5), (5–10), (10–15), (15–20)
Other	Expired

The relevant fruit categories (Apple, Banana, Carrot, and Tomato) were extracted from both datasets. Age-range labels were assigned based on the available freshness information. The Fruits-360 dataset provided age-stage labels directly, while the freshness labels from the Kaggle dataset were mapped to the project's age ranges.

An "Expired" class was created from visibly degraded specimens across the selected fruit categories. All images were then grouped into the 14 target classes to create a unified dataset with consistent labels.

Image Standardisation. The images were standardised to a consistent format. All images were converted to the RGB colour space to ensure consistent colour-channel representation. The images were resized to 224 × 224 pixels to provide a consistent input size for the MobileNetV2 classifier.

3.4.2 Data Validation

Several validation checks were performed to ensure the prepared dataset was suitable for model development.

Table 3.4: Data Validation and Quality Control Checks

Validation Check	Objective
Label Consistency	Verify target class assignment

Each image was checked to ensure it was a valid file that could be opened and processed. The colour space was verified and converted to RGB where necessary. All images were resized to 224×224 pixels. The class labels were reviewed to confirm that each image was assigned to the correct target class.

3.4.3 Data Splitting

The unified dataset was divided into three subsets for model development and evaluation.

Table 3.5: Allocation of Dataset Splits

Split	Proportion	Purpose
Test	10%	Final performance metrics

The 80/10/10 split was applied to the dataset. The training set was used to update the model weights during training. The validation set was used to monitor performance and select hyperparameters. The test set was reserved for final evaluation after training was complete.

Stratification was applied during splitting to maintain the class distribution across all three subsets. This ensured that each class was represented proportionally in the training, validation, and test sets.

A separate held-out test set of 280 images (20 images per class) was also reserved for final evaluation. This set was kept completely separate from training and validation and was not used during model development.

Note on Dataset Sizes. The combined dataset contained approximately 4,000 images after cleaning and organisation. This resulted in approximately 3,200 images for

training, 400 for validation, and 400 for testing. These numbers are approximate and reflect the data available from the two source datasets.

3.4.4 Data Augmentation

Data augmentation was applied during training to introduce realistic variation into the training images. This helps the model generalise to images captured under different conditions.

Table 3.6: Applied Data Augmentation Techniques

Technique	Parameter	Purpose
Zoom	$\pm 10\%$	Scale robustness

These augmentations were chosen to reflect the types of variation expected in real-world use. Users may hold their phone at different angles, capture images under different lighting, and frame fruits at different distances. The augmentations were applied randomly during training to increase the effective size of the dataset.

Augmentation was applied only to the training data. The validation and test sets were not augmented, as they are intended to evaluate performance on unmodified images.

3.5 Models

This section describes the three models that form the core of the Fresco system.

The models operate sequentially to transform an input fruit image into a freshness score, shelf-life estimate, and final recommendation.

3.5.1 Model 1: MobileNetV2 Convolutional Neural Network

Purpose

Model 1 is the image classification component of the system. It identifies the fruit type and its age or freshness stage from the input photograph.

Input

The model receives an RGB image resized to 224×224 pixels. The pixel values are normalised to the range $[-1, 1]$ to match the preprocessing used during the model's pre-training on ImageNet.

Architecture

The model uses the MobileNetV2 architecture as its base. MobileNetV2 was designed for efficient image classification on mobile and embedded devices (Sandler et al., 2018). It uses depthwise separable convolutions and inverted residual blocks to reduce computational cost while maintaining accuracy.

The base model was pre-trained on ImageNet, which contains 1.28 million images across 1,000 classes. A custom classification head was added on top of the base network to produce predictions for the 14 target classes used in this project.

Transfer Learning Process

The model was trained using transfer learning in two phases.

Phase 1: Frozen Base. The MobileNetV2 base layers were frozen, meaning their weights were not updated during training. Only the newly added classification head was trained. This phase ran for 10 epochs with a learning rate of 1×10^{-3} .

Phase 2: Fine-Tuning. The top 30 layers of the base network were unfrozen and trained at a lower learning rate of 1×10^{-5} for 15 epochs. This allowed the pre-trained features to be slightly adjusted to the fruit domain without causing catastrophic forgetting.

Table 3.7: Transfer Learning Training Phases

Phase	Epochs	Learning Rate	Layers Trained
Phase 2: Fine-Tuning	15	1×10^{-5}	Top 30 layers

Training Parameters

The model was trained using the Adam optimiser with $\beta_1 = 0.9$ and $\beta_2 = 0.999$.

The loss function was categorical cross-entropy, which is appropriate for multi-class classification problems.

The categorical cross-entropy loss is defined as:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i)$$

where N is the number of classes, y_i is the true label indicator, and $\hat{y}_i$ is the predicted probability for class i .

A batch size of 32 was used during training. The batch size was reduced from an initial 64 due to GPU memory constraints during the fine-tuning phase. Training was performed on Google Colab using an NVIDIA T4 GPU with 15 GB of VRAM.

The final layer uses a softmax activation function, which produces a probability distribution over the 14 target classes.

Output

Model 1 produces a predicted class label and a confidence score representing the probability of that prediction. For example, the model might output "Banana (5-10)" with a confidence of 0.91. The predicted class is passed to Model 2 for freshness scoring.

3.5.2 Model 2: Fuzzy-Logic-Inspired Freshness Mapper

Purpose

Model 2 converts the discrete age-stage prediction from Model 1 into a continuous freshness score between 0.0 and 1.0. The score represents the visual condition of the fruit, where 1.0 indicates perfectly fresh and 0.0 indicates completely spoiled.

Input

The primary input is the class label from Model 1, such as "Banana (5-10)".

Mapping Strategy

The class label is mapped to a continuous freshness score using a calibrated lookup table. The mapping was derived from the freshness annotations available in the training data and validated against human judgement.

Table 3.8: Calibration of Class to Freshness Score Ranges

Age-Range Class	Score Range	Category
Expired	0.00 – 0.19	Spoiled

The mapping can be expressed as:

$$S_{\text{freshness}} = \text{score_map}(\text{class_label})$$

where S_freshness is the resulting freshness score and score_map is the lookup structure that assigns a score value to each class label.

Example Mappings

Table 3.9: Model Mapping Logic Examples

Predicted Class	Freshness Score
Expired	0.10

Output

Model 2 produces a continuous freshness score between 0.0 and 1.0. This score is passed to Model 3 for shelf-life estimation.

3.5.3 Model 3: Multi-Factor Shelf-Life Estimator

Purpose

Model 3 estimates the remaining shelf life of the fruit in days. The estimate is calculated separately for three storage methods: room temperature, refrigerator, and freezer.

Inputs

Model 3 uses four inputs:

1. Fruit identity from Model 1
2. Freshness score from Model 2
3. Storage method selected by the user
4. Fruit reference data from the database

The reference data includes base shelf-life values for each storage method, the fruit's optimal temperature range, and other relevant information.

Main Equation

The shelf-life estimate is calculated using the following equation:

$$E_{\text{days}} = B \times S^{1.5} \times T_f \times R_f$$

where:

- E_{days} = estimated remaining shelf life in days
- B = base shelf life for the fruit and storage method
- S = freshness score from Model 2
- T_f = temperature factor
- R_f = ripeness factor

Temperature Factor

The temperature factor adjusts the estimate based on how well the storage temperature matches the fruit's optimal temperature range. Storage temperatures are fixed at 22°C for room temperature, 4°C for refrigerator, and -18°C for freezer.

When the storage temperature falls within the optimal range, the factor is 1.0:

$$T_f = 1.0$$

$$T_f = 1.0$$

When the temperature is outside the optimal range, a penalty is applied:

$$T_f = \max(1.0 - 0.02d, 0.5)$$

$$T_f = \max(1.0 - 0.02d, 0.5)$$

where d is the deviation from the optimal temperature range in degrees Celsius.

For freezer storage, the factor is fixed at:

$$T_f = 0.95$$

This reflects the small quality loss associated with freezing and thawing.

Ripeness Factor

The ripeness factor adjusts the estimate based on the fruit's current ripening stage, which is derived from the freshness score.

Table 3.10: Ripeness Multipliers for Shelf-Life Estimation

Freshness Score	Ripeness Stage	Multiplier
< 0.50	Overripe	0.30

Worked Example

To illustrate the calculation, consider a banana with a freshness score of 0.72 stored in the refrigerator.

The fruit reference data for banana gives a refrigerator-based shelf life of 7 days and an optimal temperature range of 13°C to 15°C.

The freshness factor is:

$$S_{1.5} = 0.72_{1.5} = 0.611$$

The temperature factor for refrigerator storage (4°C) is 0.82, as the temperature is below the optimal range.

The freshness score of 0.72 falls in the Nearly Ripe category, giving a ripeness factor of 0.85.

The estimated shelf life is:

$$E_{\text{days}} = 7 \times 0.611 \times 0.82 \times 0.85 = 2.98 \text{ days}$$

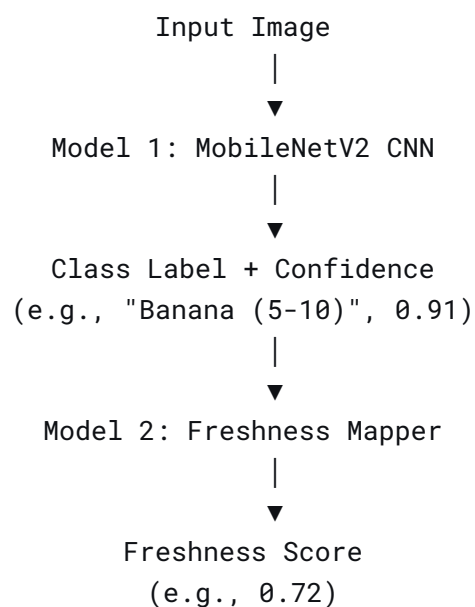
Output

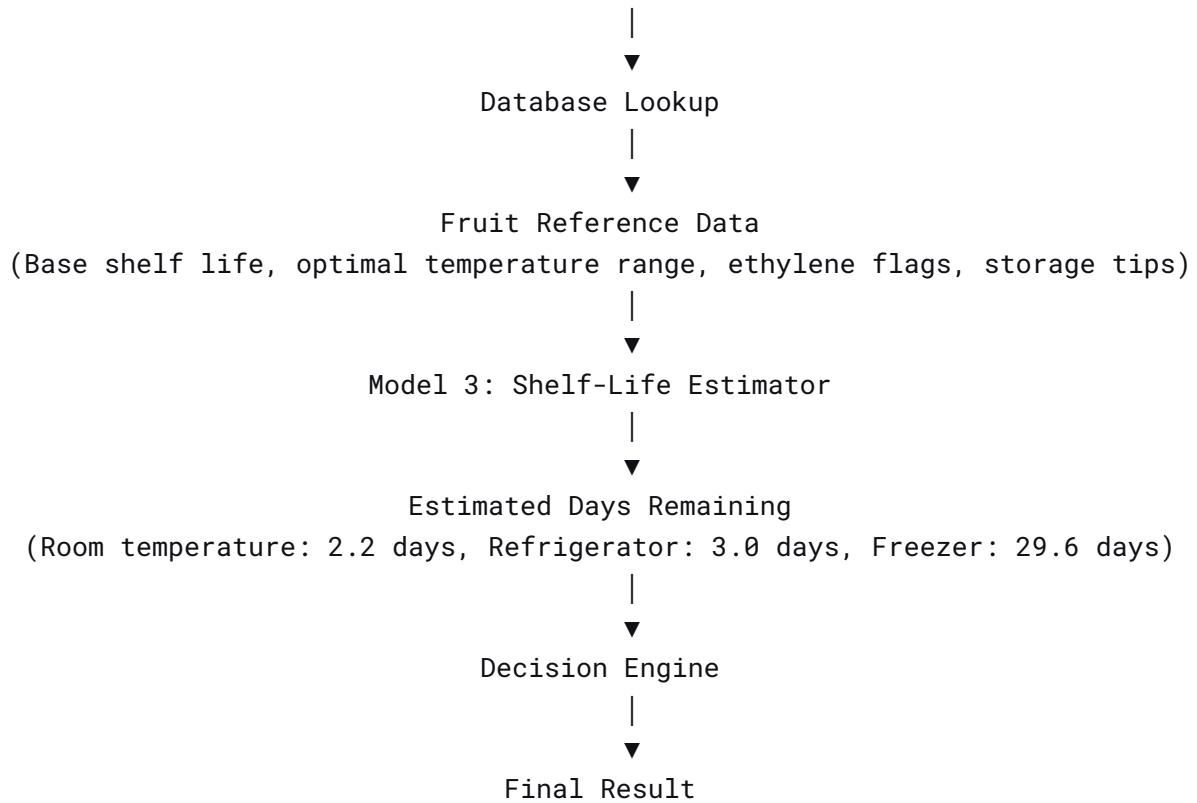
Model 3 produces estimated days remaining for each of the three storage methods. These estimates are passed to the decision engine for final verdict generation.

3.5.4 Model Interaction and Processing Pipeline

The three models operate as an integrated pipeline. Each model receives input from the previous stage and passes its output to the next stage.

Integration of the Fresco Modelling Pipeline





(Status, days remaining, recommended storage, recommendation string, ethylene advisory)

The pipeline begins when the user submits an image. Model 1 classifies the image and produces a class label and confidence score. Model 2 maps the class label to a continuous freshness score. A database lookup retrieves the fruit's reference data. Model 3 uses the freshness score and reference data to estimate shelf life for each storage method. The decision engine then combines these outputs to generate the final verdict and recommendation.

The decision engine uses a composite score that combines the freshness score and the normalised days remaining:

$$C=(S\times0.6)+(D_{\text{norm}}\times0.4)$$

where S is the freshness score and D_{norm} is the normalised days remaining, calculated as the ratio of estimated days to the base shelf life. The composite score is mapped to one of four statuses: FRESH, EAT_SOON, EAT_TODAY, or SPOILED. The recommended storage method is the one that gives the longest remaining shelf life.

The final result includes the status, estimated days remaining, recommended storage method, a recommendation string, and an ethylene advisory note where applicable.

3.6 Determination of Freshness Score (0–1)

This section explains how the system converts the predicted age-stage class into a numerical freshness score, how that score is interpreted, and how it contributes to the final decision.

3.6.1 Freshness Score Calculation

The freshness score is generated by Model 2, the fuzzy-logic-inspired freshness mapper. The process begins with the class label predicted by Model 1, such as "Banana (5–10)". This class label is passed to a mapping function that assigns a value between 0.0 and 1.0.

The mapping is expressed as:

$$S_{\text{freshness}} = \text{score_map}(\text{class_label})$$

where:

- $S_{\text{freshness}}$: is the resulting freshness score
- `class_label` : is the predicted age-stage class from Model 1
- `score_map` : is a calibrated lookup table that assigns a score to each class

A score close to 1.0 indicates a fresher condition, while a score close to 0.0 indicates a more deteriorated condition.

Example Mappings

Table 3.11: Examples of Numerical Score Calibration

Predicted Class	Freshness Score
Expired	0.10

These values illustrate how different predicted classes map to different freshness scores.

3.6.2 Freshness Category Mapping

The continuous 0–1 score is grouped into five categories to make the result easier for users to understand. Each category represents a general freshness condition and provides guidance on how the fruit should be handled.

Table 3.12: Interpretation of Freshness Categories

Score Range	Category	Interpretation
0.00 – 0.19	Spoiled	No meaningful shelf life

The category is derived from the numerical score. The score is calculated first, and the category is then assigned based on the range in which the score falls.

3.6.3 Composite Decision Score

The freshness score is one of the inputs used by the Decision Engine to generate the final status of the fruit. The Decision Engine combines the freshness score with the estimated remaining shelf life to produce a composite decision score.

The composite score is calculated as:

$$C = (S_{\text{freshness}} \times 0.6) + (D_{\text{norm}} \times 0.4)$$

where:

- C is the composite decision score
- $S_{\text{freshness}}$ is the freshness score from Model 2

- D_{norm} is the normalised remaining shelf life

The normalised remaining shelf life is calculated as:

$$D_{\text{norm}} = \frac{\text{Days Remaining}}{\text{Base Shelf Life}}$$

$$D_{\text{norm}} = \frac{\text{Days Remaining}}{\text{Base Shelf Life}}$$

The value is constrained to the range 0 to 1.

The composite score combines two sources of information. The freshness score, weighted at 60%, represents the visual condition of the fruit. The normalised remaining shelf life, weighted at 40%, represents how much of the expected shelf life remains relative to the baseline.

The composite score is then mapped to one of four statuses:

Table 3.13: Composite Score to Status Mapping and Actions

Composite Score (C)	Status	Action
< 0.20	SPOILED	Discard

The sequence from prediction to final status is:

Predicted Class → Freshness Score → Shelf-Life Estimate → Normalised Days →
Composite Score → Final Status

This process ensures that both the visual assessment of the fruit and its estimated remaining shelf life contribute to the final recommendation provided to the user.

3.7 Deployment

This section describes how the Fresco system was assembled as a working application and how a user request moves through the deployed components to produce a result.

3.7.1 Deployed System Components

The Fresco system was deployed as a web-based application consisting of a frontend, a backend, a database, and a model serving component.

Frontend: The frontend was implemented as a Next.js Progressive Web App. Users access the application through a mobile browser and can install it on their device's home screen. The frontend provides the user interface for capturing or uploading images, viewing scan results, managing inventory, and checking storage compatibility. It communicates with the backend through HTTP requests.

Backend: The backend was implemented using FastAPI. It receives requests from the frontend, validates input, runs the scan processing pipeline, and returns structured results. The backend exposes API endpoints for authentication, scanning, inventory management, and fruit reference queries. It uses asynchronous request handling to support concurrent requests.

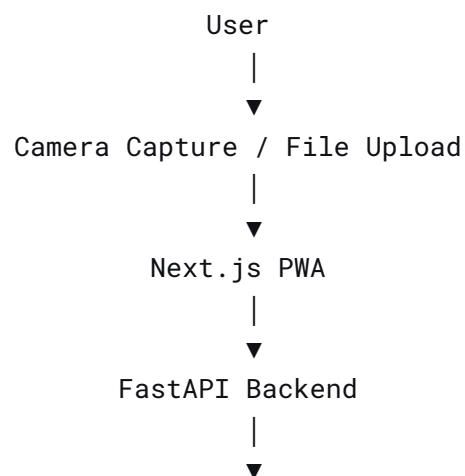
Database: The deployed system uses PostgreSQL in production and MySQL during development. The database stores the fruit reference catalogue, user accounts, personal inventory records, and scheduled notifications. The backend queries the database during the scan process to retrieve reference data and updates it when users add or modify inventory items.

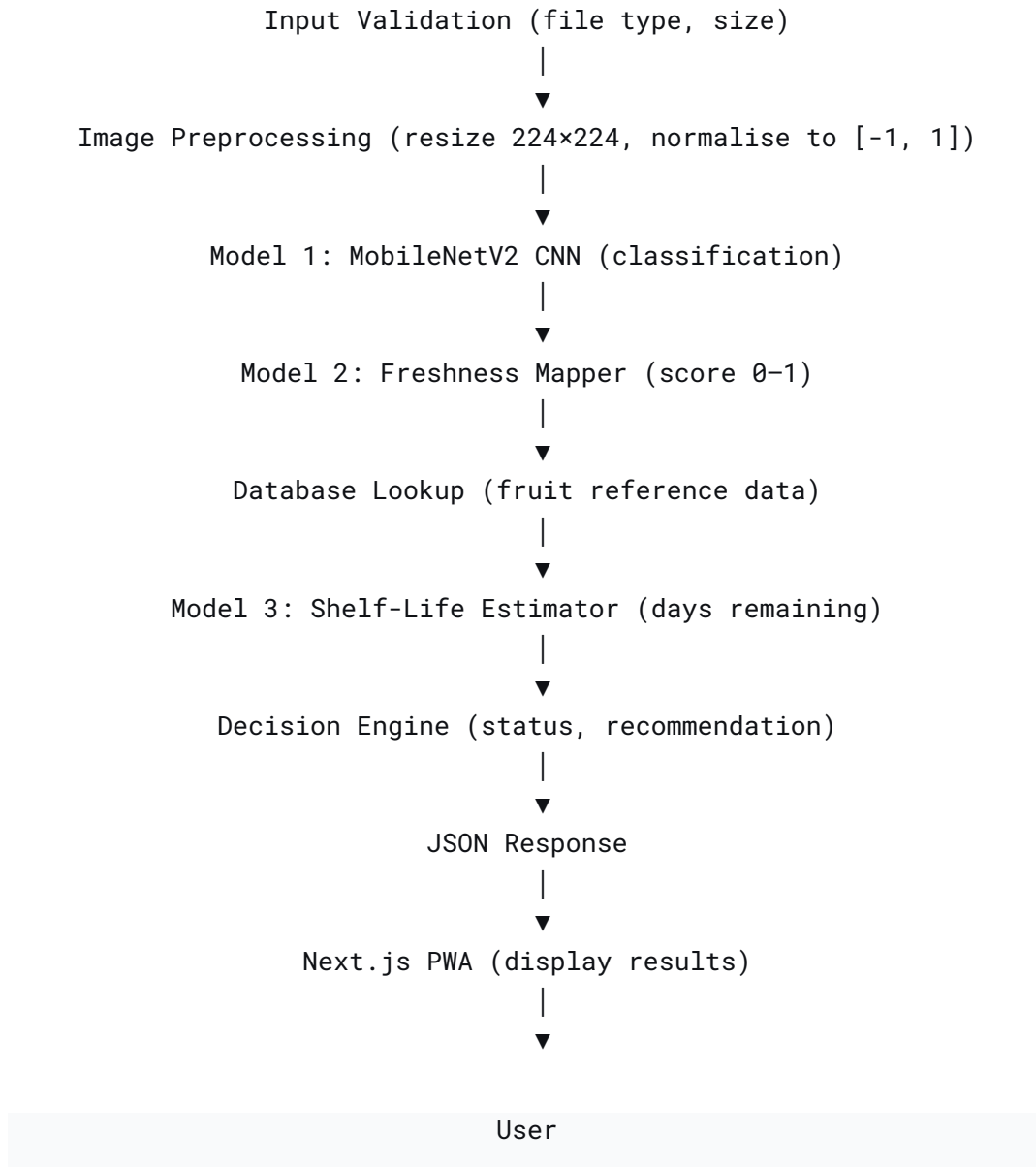
Model Serving: The trained MobileNetV2 model is loaded into memory when the application starts. A warm-up prediction is performed using dummy input to initialise the computation graph. The model is then retained in memory for the lifetime of the server process, so it does not need to be reloaded for each scan request.

3.7.2 Request and Response Flow

The complete flow of a scan request through the deployed system is shown below:

Request and Response Flow of the Deployed Fresco System





The user captures an image using the device camera or selects an existing image. The frontend sends the image to the backend through the scan API endpoint as multipart form data.

The backend validates the uploaded file by checking the file type and size. The image is then preprocessed by resizing it to 224×224 pixels and normalising the pixel values to the $[-1, 1]$ range expected by the model.

Model 1, the MobileNetV2 classifier, analyses the preprocessed image and produces a predicted class label and confidence score. Model 2, the freshness mapper, converts the class label into a continuous freshness score between 0 and 1.

The backend queries the database to retrieve the fruit's reference data, including base shelf-life values, optimal temperature range, and ethylene flags. Model 3, the shelf-life estimator, uses the freshness score and reference data to calculate estimated days remaining for room temperature, refrigerator, and freezer storage.

The decision engine combines the freshness score and normalised days remaining into a composite score, maps it to a status, and generates a recommendation string. Where applicable, an ethylene compatibility check is performed against the user's existing inventory.

The backend returns the complete result as a structured JSON response. The frontend displays the result to the user, including the status, estimated days remaining, recommended storage method, recommendation message, and ethylene advisory note where applicable. The user can then choose to save the scanned fruit to their personal inventory.

3.7.3 Deployment and Performance Characteristics

The main deployment and performance characteristics of the system are summarised in Table 3.6.

Table 3.14: Deployment and Performance Characteristics

Metric	Value	Description
Supported formats	JPEG, PNG, WebP	Accepted image formats

The inference latency of 200 to 500 milliseconds is measured after the model has been loaded and warmed up. The first request after server startup takes approximately 4 to 5 seconds due to the initialisation of the model's computation graph.

The backend is configured to handle concurrent requests through asynchronous request handling, with the model inference executed in a thread pool to avoid blocking other requests. Cross-origin requests are restricted to configured application origins.